

WORK EXPERIENCE

- > **Associate Software Engineer @ Fidelity Investments:** Fund Operations Platform Maintenance *June 2025 – Now*
 - Performed modernization and maintenance across over 80 SpringBoot applications.
 - Developed a workflow using OpenRewrite that halved the time spent on SpringBoot major version upgrades.
 - Expanded Cucumber end-to-end tests to be able to test applications requiring Active Directory authentication.
 - Developed automation using OpenRewrite, SonarQube CLI, and Github Copilot to fix code quality issues.
 - Introduced use of OpenApi code generation to reduce development time and bugs.
 - Followed Agile methodology and test-driven development with tools like Jira, Jasmine, Cypress, and JUnit.
- > **Computational Protein Design Co-op @ Tessera Therapeutics:** Gene Therapy *Jan. 2024 – June 2024*
 - Used statistical techniques to test biological hypothesis from protein engineering data.
 - Built an internal web app to expedite and democratize protein design analysis using Flask and Plotly.
 - Integrated molecular structure libraries and custom plot configurations to enable interactive visualizations.
- > **Software Engineering Co-op @ Nth Cycle:** Metal Refining Startup *Jan. 2023 – June 2023*
 - Designed and implemented a React webapp (see /lucas-demo-screenshots) which used a drag and drop library for no-code design of forms, pipelines, and charts. Deployed to MongoDB Atlas and Microsoft Azure.
 - Fixed problems with internet, printers, laptops, AzureVPN and Remote Desktop in lieu of an IT department.
- > **Peer Tutoring:** Discrete, CS 1 & 2, OOD, Algo, Phys 2, Calc 3, LinAlg, Prob & Stats *Sep. 2022 – June 2023*
 - Demonstrate patience and professionalism with tutees having a variety of skill levels.
 - 164 hours total $\approx$ 4 hrs/wk.
- > **Sijia Dong Lab:** Computational chemistry research group *Dec. 2021 – Jan. 2023*
 -  Designed a QtPy GUI and script to autogenerate protein-ligand docking poses in Schrodinger Maestro.
 -  Used Bash, MATLAB, Python, and TensorFlow to transform molecular coordinates into a normalized form using matrix manipulations for use in an explainable AI model to understand how different ligands influence the excitation wavelengths and strengths in different docking poses.
- > **Internship with Town of Holliston Director of Technology:** *Nov. 2020 – Feb. 2021, Apr. 2021 – May 2021*
 - Setup and resolve problems with phones, printers, desktops, software, and file storage for municipal departments.

EDUCATION

- > **Northeastern University** | Khoury College of Computer Sciences | Honors Boston, MA
B.S. Computer Science & Mathematics \ \ Physics & Chemistry Minor \ \ 3.79 GPA *2021 – 2025*

COURSES AND TECH STACK

- > **Tech:** JavaScript/TypeScript (Angular/React), Java/SpringBoot, Python, Bash/Linux, Lean, Racket, Rust
- > **CS:** Software Dev, AI, Networks & Distributed Systems, Object Oriented Design, Logic & Computation
- > **Math:** PDEs, Statistics & Stochastic Processes, Linear Algebra, Number Theory, Group Theory, Calculus 3
- > **Chemistry:** Organic Chemistry 1 & 2, Physical Chemistry, Analytical Chemistry, Quantum Chemistry
- > **Physics:** Electronics, Quantum Computation and Information, Modern Physics

PROJECTS

- CA-DSL** | *Typed Racket*  /zedeckj/ca-dsl *Spring 2025*
Co-authored a domain specific language for describing and executing variants of Conway's Game of Life as part of a course in my final semester. Made extensive use of Racket's macro system for terser and more readable syntax, and also made extensive use of parametric polymorphism to maximize extensibility to different variants.

ONGOING LEARNING

- Programming Languages** |  A course than ran after my graduation examining the PL design choices behind JS/TS.
- Probability & Statistics** |  More advanced study of probability and stochastic processes grounded in measure theory
- Type Theory** |  Learned some type theory from a textbook which uses Lean to make the theory more concrete.